

UltraClear 4K Smart Microscope System

Precision Imaging Meets Intelligent Automation



**Intelligent Optical Imaging.
Automated Microscopy Control.
Built for Research & Inspection.**

The UltraClear 4K Smart Microscope System is a high-performance hybrid imaging platform engineered for advanced scientific research, semiconductor inspection, metallurgical analysis, and precision industrial microscopy.

Combining apochromatic optics, intelligent automation, 4K ultra-HD imaging, and integrated measurement capabilities, the system delivers exceptional clarity, true-color fidelity, and high-resolution visualization across demanding inspection workflows.

Designed as a modular and future-ready microscopy platform, UltraClear seamlessly bridges traditional optical microscopy with modern digital imaging and AI-assisted analysis.

KEY FEATURES

● Apochromatic Imaging System

High-performance apochromatic optics eliminate focus shift across wavelengths while delivering true-color fidelity, high contrast, and reduced chromatic aberration.

● 4K Ultra-HD Imaging System

Sony CMOS sensor enables 3840 × 2160 @ 60 fps ultra-low-noise real-time imaging with HDMI 2.0 output, C-Mount compatibility, and local/network storage support.

● 10:1 Intelligent Zoom Architecture

0.3X–3.0X zoom system with manual or motorized options provides ultra-low distortion and up to 1.5× larger field of view.

● Smart Measurement

Supports measurement of lines, circles, angles, and point-to-line distances with automatic edge detection, real-time overlays, and export to PDF, CSV, and reports.

● Infinity Plan Semi-APO Objectives

20X / 50X objectives with 0.4 / 0.55 NA and 8–10 mm working distance provide flexible high-precision imaging.

Notes

Optical performance depends on selected objective and illumination configuration

Modular upgrades available for advanced imaging workflows

Compatible with laboratory and industrial inspection environments

AI imaging capabilities may vary by software configuration

APPLICATIONS

- Semiconductor inspection
- Metallurgical failure analysis
- Biological tissue observation
- PCB and electronics testing
- Biomedical research
- Academic laboratories and R&D environments
- 2D Materials; optical chips & material inspection



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SPECIFICATIONS

OPTICAL SYSTEM

Optical Type	Apochromatic
Zoom Ratio	10:1
Zoom Range	0.3X–3.0X
Optical Path	Ultra-low distortion parallel path
Field of View	Up to 1.5× wider than conventional systems

OBJECTIVE SYSTEM

Objective Type	Infinity Plan Semi-APO
Magnification Options	20X / 50X
Numerical Aperture	0.4 / 0.55
Working Distance	8–10 mm

IMAGING SYSTEM

Sensor Type	Sony CMOS
Resolution	3840 × 2160
Frame Rate	60 fps
Output Interface	HDMI 2.0
Mount Type	C-Mount

ILLUMINATION SYSTEM

Illumination Type	Epi-illumination
Light Source	Precision LED
Polarization Imaging	Supported
Intensity Control	Adjustable

SOFTWARE & PROCESSING

Measurement Tools	Integrated real-time metrology
Functions	Edge and contour detection
Data Export	PDF / CSV / reports

MECHANICAL CONSTRUCTION

Body Material	CNC aluminum alloy
Surface Finish	Anodized
Focus Mechanism	Precision manual control
Design Type	Modular and upgradeable

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